ICML

2026

REPRODUCTION

Learning U-Statistics with Active Inference: Reproduction Audit

CPU

AUDIT

OPEN

Source-pinned CPU evidence separates scoped mechanism checks, a comparator falsification, and unavailable empirical artifacts.

DineshAI reproduction campaign

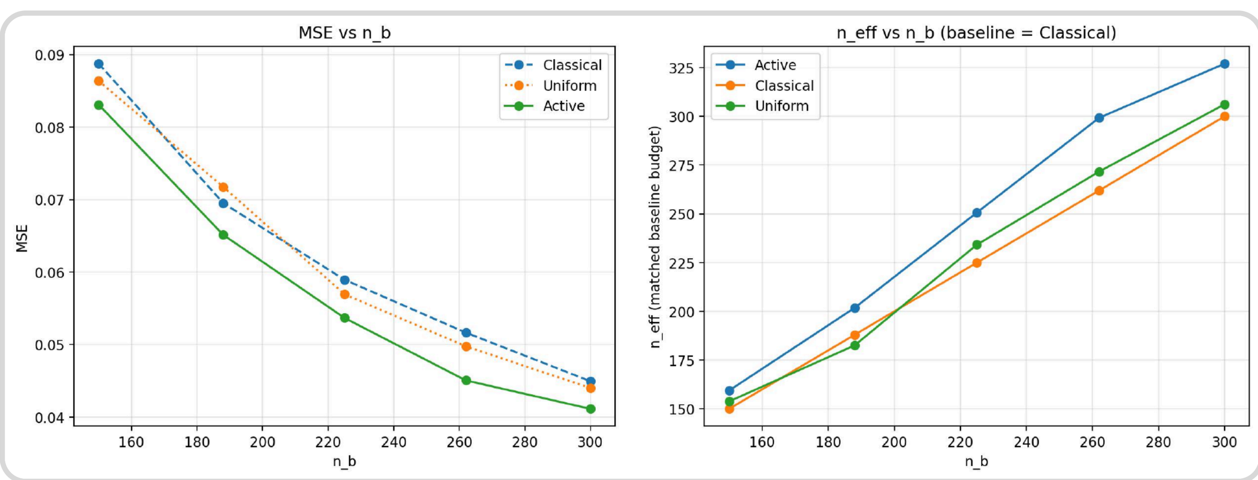
Independent reproduction record

1 Outcome

Six live claims were audited against the pinned paper source and official release. Three finite mechanism audits are scoped verified; one literal comparator claim is falsified; two empirical claims are inconclusive.

2 AIPW unbiasedness

64-cell exact enumeration gives AIPW error 1.11e-16; omitting IPW is biased by 0.0026. This finite exact check supports the mechanism.



3 Optimal sampling

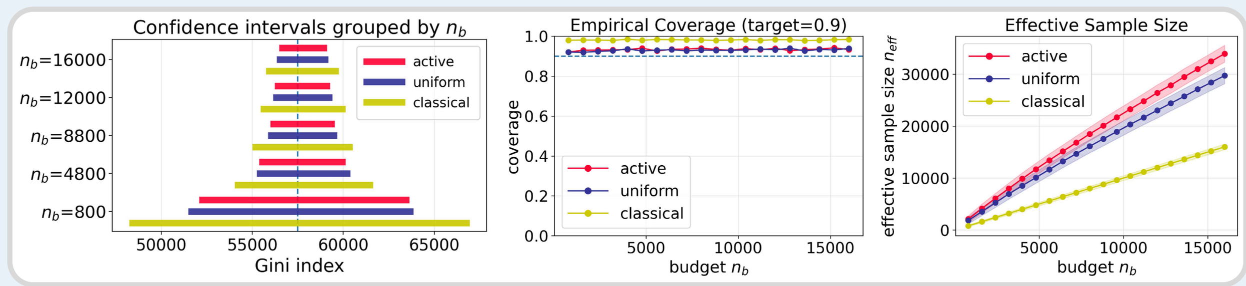
The source projection-residual sqrt(s) allocation has objective 0.62886 versus 0.65652 for a raw-residual control.

4 CRN coupling

CRN coupling RMS shrinks from 0.25454 to 0.12384 under H2; a fixed-error control stays near 0.95.

5 ACS comparator

The live ≈60% ACS saving is attributed to uniform AIPW, but the source attributes it to classical IPW.



6 VitalDB

Source supports the 20%-versus-classical VitalDB wording, and a source-faithful rerun remains unavailable.

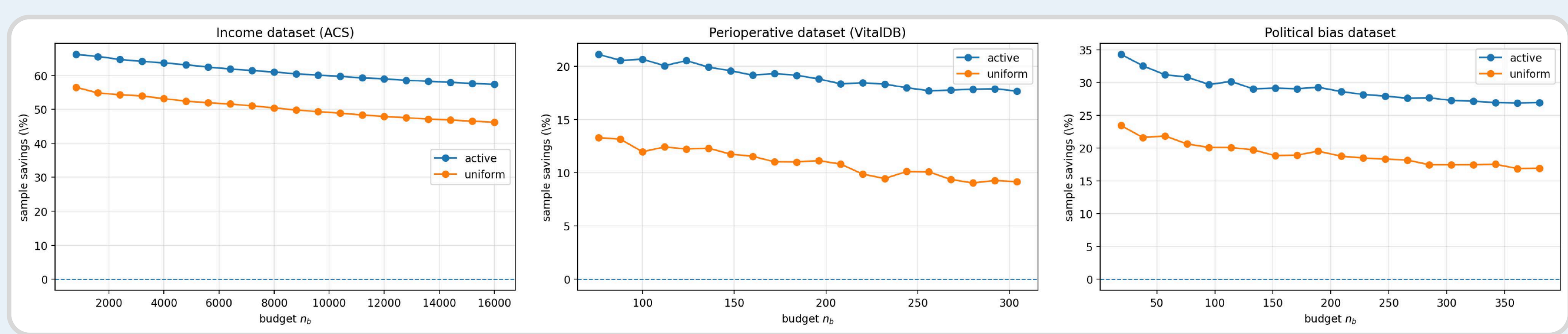
7 Political-bias labels

Source supports 20%-versus-classical and 10%-versus-uniform political-bias savings, because the source did not release its data or LLM outputs.

8 Reproducibility

All scripts, tests, source pins, raw logs, and hashes are public. No HF Job or Bucket was used; unreleased empirical artifacts are documented rather than replaced with proxies.

- Public code at the pinned commit.
- Exact live claims retained with hashes.



Active U-statistics reproduction audit · CPU-only · source-pinned

Code: **github.com/MachineLearning-Nerd/icml26-repro-BwufLjXbMO-learning-u-statistics-active-inference**